Practical 3

Object Oriented programs in C#, Programs on delegates and events.

Q. 1-a Write a C# .NET Core program to create a class named `Student` with data members Roll Number and Name. Create an object of the class and display the student information.

```
using System;
namespace StudentDemo
{
   class Student
   {
      public int RollNo;
      public string Name;

      public void Display()
      {
         Console.WriteLine("Roll No: " + RollNo);
         Console.WriteLine("Name: " + Name);
      }
   }

   class Program
   {
      static void Main(string[] args)
      {
         Student s = new Student();

         s.RollNo = 101;
         s.Name = "Rahul";

         s.Display();

         Console.ReadLine();
      }
   }
}
```

Constructor

Q. 1-b. Write a program to demonstrate the use of a constructor for initializing employee information.

```
using System;

namespace ConstructorDemo
{
   class Employee
   {
```

```
        int empId;
        string empName;

        public Employee(int id, string name)
        {
            empId = id;
            empName = name;
        }

        public void Display()
        {
            Console.WriteLine("Employee ID: " + empId);
            Console.WriteLine("Employee Name: " + empName);
        }
    }

    class Program
    {
        static void Main(string[] args)
        {
            Employee e = new Employee(1, "Priya");
            e.Display();

            Console.ReadLine();
        }
    }
}
```

Inheritance

Q. 1-c. Write a program to demonstrate single inheritance by creating a `Person` class and deriving a `Student` class from it.

```
using System;

namespace SingleInheritance
{
    class Person
    {
        public string Name = "Amit";
    }

    class Student : Person
    {
        public void Display()
        {
            Console.WriteLine("Student Name: " + Name);
        }
    }
```

```
    class Program
    {
        static void Main(string[] args)
        {
            Student s = new Student();
            s.Display();

            Console.ReadLine();
        }
    }
}
```

Method Overloading

Q. 1-d. Write a program to demonstrate method overloading by creating multiple methods to calculate the sum of numbers.

```
using System;

namespace MethodOverloading
{
    class Addition
    {
        public int Sum(int a, int b)
        {
            return a + b;
        }

        public int Sum(int a, int b, int c)
        {
            return a + b + c;
        }
    }

    class Program
    {
        static void Main(string[] args)
        {
            Addition obj = new Addition();

            Console.WriteLine("Sum = " + obj.Sum(10, 20));
            Console.WriteLine("Sum = " + obj.Sum(10, 20, 30));

            Console.ReadLine();
        }
    }
}
```

Interface

Q. 1-e.Write a program to demonstrate the use of an interface.

```
using System;

namespace InterfaceDemo
{
   interface IPrinter
   {
      void Print();
   }

   class Document : IPrinter
   {
      public void Print()
      {
         Console.WriteLine("Document is printing...");
      }
   }

   class Program
   {
      static void Main(string[] args)
      {
         Document d = new Document();
         d.Print();

         Console.ReadLine();
      }
   }
}
```

Program on Delegates

Pract 2. Write a program to create a delegate that displays a welcome message.

```
using System;

namespace DelegateDemo
{
   delegate void Message();

   class Program
   {
      static void Welcome()
      {
         Console.WriteLine("Welcome to C# Delegates!");
      }

      static void Main(string[] args)
      {
         Message msg = Welcome;
```

```
            msg();
            Console.ReadLine();
        }
    }
}
```

Programs on Events

Practical 3. Write a program to create an event that displays a message when a button click event occurs.

```
using System;

namespace EventDemo
{
    class Button
    {
        public delegate void ClickHandler();

        public event ClickHandler Click;

        public void OnClick()
        {
            if (Click != null)
                Click();
        }
    }

    class Program
    {
        static void Display()
        {
            Console.WriteLine("Button Clicked!");
        }

        static void Main(string[] args)
        {
            Button b = new Button();
            b.Click += Display;
            b.OnClick();
            Console.ReadLine();
        }
    }
}
```